

Imię i nazwisko: Paweł Borowiecki

Proces budowania pakietu GNU Hello ze źródeł:

Na początku należy odkomentować linie pliku `/etc/apt/sources.list`, zawierające „deb-src”.

```
paweł@myubuntu:~$ sudo nano /etc/apt/sources.list
```

```
GNU nano 6.2 /etc/apt/sources.list *
deb cdrom:[Ubuntu 22.04.3 LTS _Jammy Jellyfish_ - Release amd64 (20230807.2)]/ jammy main restricted

# See http://help.ubuntu.com/community/UpgradeNotes for how to upgrade to
# newer versions of the distribution.
deb http://pl.archive.ubuntu.com/ubuntu/ jammy main restricted
deb-src http://pl.archive.ubuntu.com/ubuntu/ jammy main restricted

## Major bug fix updates produced after the final release of the
## distribution.
deb http://pl.archive.ubuntu.com/ubuntu/ jammy-updates main restricted
deb-src http://pl.archive.ubuntu.com/ubuntu/ jammy-updates main restricted

## N.B. software from this repository is ENTIRELY UNSUPPORTED by the Ubuntu
## team. Also, please note that software in universe WILL NOT receive any
## review or updates from the Ubuntu security team.
deb http://pl.archive.ubuntu.com/ubuntu/ jammy universe
deb-src http://pl.archive.ubuntu.com/ubuntu/ jammy universe
deb http://pl.archive.ubuntu.com/ubuntu/ jammy-updates universe
deb-src http://pl.archive.ubuntu.com/ubuntu/ jammy-updates universe

## N.B. software from this repository is ENTIRELY UNSUPPORTED by the Ubuntu
## team, and may not be under a free licence. Please satisfy yourself as to
## your rights to use the software. Also, please note that software in
## multiverse WILL NOT receive any review or updates from the Ubuntu
## security team.
deb http://pl.archive.ubuntu.com/ubuntu/ jammy multiverse
deb-src http://pl.archive.ubuntu.com/ubuntu/ jammy multiverse
deb http://pl.archive.ubuntu.com/ubuntu/ jammy-updates multiverse
deb-src http://pl.archive.ubuntu.com/ubuntu/ jammy-updates multiverse

## N.B. software from this repository may not have been tested as
## extensively as that contained in the main release, although it includes
## newer versions of some applications which may provide useful features.
## Also, please note that software in backports WILL NOT receive any review
## or updates from the Ubuntu security team.
```

```

GNU nano 6.2 /etc/apt/sources.list *
deb-src http://pl.archive.ubuntu.com/ubuntu/ jammy universe
deb http://pl.archive.ubuntu.com/ubuntu/ jammy-updates universe
deb-src http://pl.archive.ubuntu.com/ubuntu/ jammy-updates universe

## N.B. software from this repository is ENTIRELY UNSUPPORTED by the Ubuntu
## team, and may not be under a free licence. Please satisfy yourself as to
## your rights to use the software. Also, please note that software in
## multiverse WILL NOT receive any review or updates from the Ubuntu
## security team.
deb http://pl.archive.ubuntu.com/ubuntu/ jammy multiverse
deb-src http://pl.archive.ubuntu.com/ubuntu/ jammy multiverse
deb http://pl.archive.ubuntu.com/ubuntu/ jammy-updates multiverse
deb-src http://pl.archive.ubuntu.com/ubuntu/ jammy-updates multiverse

## N.B. software from this repository may not have been tested as
## extensively as that contained in the main release, although it includes
## newer versions of some applications which may provide useful features.
## Also, please note that software in backports WILL NOT receive any review
## or updates from the Ubuntu security team.
deb http://pl.archive.ubuntu.com/ubuntu/ jammy-backports main restricted universe multiverse
deb-src http://pl.archive.ubuntu.com/ubuntu/ jammy-backports main restricted universe multiverse

deb http://security.ubuntu.com/ubuntu jammy-security main restricted
deb-src http://security.ubuntu.com/ubuntu jammy-security main restricted
deb http://security.ubuntu.com/ubuntu jammy-security universe
deb-src http://security.ubuntu.com/ubuntu jammy-security universe
deb http://security.ubuntu.com/ubuntu jammy-security multiverse
deb-src http://security.ubuntu.com/ubuntu jammy-security multiverse

# This system was installed using small removable media
# (e.g. netinst, live or single CD). The matching "deb cdrom"
# entries were disabled at the end of the installation process.
# For information about how to configure apt package sources,
# see the sources.list(5) manual.

```

Następnie należy pobrać najnowsze informacje o dostępnych pakietach, za pomocą polecenia `sudo apt update` i pobrać plik źródłowe z serwera lustrzanego, za pomocą `apt source hello`.

```

pawel@myubuntu:~$ sudo apt update
Stary:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Stary:2 http://pl.archive.ubuntu.com/ubuntu jammy InRelease
Stary:3 https://download.docker.com/linux/ubuntu jammy InRelease
Stary:4 http://pl.archive.ubuntu.com/ubuntu jammy-updates InRelease
Stary:5 http://pl.archive.ubuntu.com/ubuntu jammy-backports InRelease
Pobieranie:6 http://security.ubuntu.com/ubuntu jammy-security/multiverse Sources [23,9 kB]
Pobieranie:7 http://security.ubuntu.com/ubuntu jammy-security/restricted Sources [71,4 kB]
Pobieranie:8 http://security.ubuntu.com/ubuntu jammy-security/universe Sources [233 kB]
Pobieranie:9 http://security.ubuntu.com/ubuntu jammy-security/main Sources [266 kB]
Pobieranie:10 http://pl.archive.ubuntu.com/ubuntu jammy/restricted Sources [23,7 kB]
Pobieranie:11 http://pl.archive.ubuntu.com/ubuntu jammy/main Sources [1 340 kB]
Pobieranie:12 http://pl.archive.ubuntu.com/ubuntu jammy/universe Sources [17,8 MB]
Pobieranie:13 http://pl.archive.ubuntu.com/ubuntu jammy/multiverse Sources [304 kB]
Pobieranie:14 http://pl.archive.ubuntu.com/ubuntu jammy-updates/multiverse Sources [31,2 kB]
Pobieranie:15 http://pl.archive.ubuntu.com/ubuntu jammy-updates/restricted Sources [75,0 kB]
Pobieranie:16 http://pl.archive.ubuntu.com/ubuntu jammy-updates/universe Sources [343 kB]
Pobieranie:17 http://pl.archive.ubuntu.com/ubuntu jammy-updates/main Sources [423 kB]
Pobieranie:18 http://pl.archive.ubuntu.com/ubuntu jammy-backports/universe Sources [9 796 B]
Pobieranie:19 http://pl.archive.ubuntu.com/ubuntu jammy-backports/main Sources [7 876 B]
Pobrano 21,0 MB w 3s (6 091 kB/s)
Czytanie list pakietów... Gotowe
Budowanie drzewa zależności... Gotowe
Odczyt informacji o stanie... Gotowe
14 pakietów może być zaktualizowanych. Można je zobaczyć wykonując 'apt list --upgradable'.
pawel@myubuntu:~$ apt source hello
Czytanie list pakietów... Gotowe
Konieczne pobranie 734 kB archiwów źródeł.
Pobieranie:1 http://pl.archive.ubuntu.com/ubuntu jammy/main hello 2.10-2ubuntu4 (dsc) [1 465 B]
Pobieranie:2 http://pl.archive.ubuntu.com/ubuntu jammy/main hello 2.10-2ubuntu4 (tar) [726 kB]
Pobieranie:3 http://pl.archive.ubuntu.com/ubuntu jammy/main hello 2.10-2ubuntu4 (diff) [6 832 B]
Pobrano 734 kB w 1s (670 kB/s)
dpkg-source: info: rozpakowywanie hello w hello-2.10
dpkg-source: info: rozpakowywanie hello_2.10.orig.tar.gz
dpkg-source: info: rozpakowywanie hello_2.10-2ubuntu4.debian.tar.xz
pawel@myubuntu:~$

```

Potem należy przejść do rozpakowanego katalogu i pobrać zależności, za pomocą komendy `sudo apt-get build-dep hello`.

```
pawel@myubuntu:~$ cd hello-*
pawel@myubuntu:~/hello-2.10$ sudo apt-get build-dep hello
Czytanie list pakietów... Gotowe
Czytanie list pakietów... Gotowe
Budowanie drzewa zależności... Gotowe
Odczyt informacji o stanie... Gotowe
Następujące pakiety zostały zainstalowane automatycznie i nie są już więcej wymagane:
  bridge-utils ubuntu-fan
Aby je usunąć należy użyć "sudo apt autoremove".
0 aktualizowanych, 0 nowo instalowanych, 0 usuwanych i 14 nieaktualizowanych.
pawel@myubuntu:~/hello-2.10$
```

Kolejnym krokiem jest zbudowanie pakietu, za pomocą polecenia `sudo debuild -b -uc -us`.

```
pawel@myubuntu:~/hello-2.10$ sudo debuild -b -uc -us
dpkg-buildpackage -us -uc -ui -b
dpkg-buildpackage: info: pakiet źródłowy hello
dpkg-buildpackage: info: wersja pakietu źródłowego 2.10-2ubuntu4
dpkg-buildpackage: info: dystrybucja źródła jammy
dpkg-buildpackage: info: źródła zmienione przez Łukasz 'sil2100' Zemczak <lukasz.zemczak@ubuntu.com>
dpkg-source --before-build .
dpkg-buildpackage: info: architektura gościa amd64
debian/rules clean
dh clean
dh: warning: Compatibility levels before 10 are deprecated (level 9 in use)
  debian/rules override_dh_auto_clean
make[1]: Wejście do katalogu '/home/pawel/hello-2.10'
[ ! -f Makefile ] || /usr/bin/make distclean
make[1]: Opuszczenie katalogu '/home/pawel/hello-2.10'
dh_clean
dh_clean: warning: Compatibility levels before 10 are deprecated (level 9 in use)
  debian/rules binary
dh binary
dh: warning: Compatibility levels before 10 are deprecated (level 9 in use)
  dh_update_autotools_config
  dh_auto_configure
dh_auto_configure: warning: Compatibility levels before 10 are deprecated (level 9 in use)
./configure --build=x86_64-linux-gnu --prefix=/usr --includedir=${prefix}/include --mandir=${prefix}/share/man --infodir=${prefix}/share/info --sysconfdir=/etc --localstatedir=/var --disable-option-checking --disable-silent-rules --libdir=${prefix}/lib/x86_64-linux-gnu --libexecdir=${prefix}/lib/x86_64-linux-gnu --disable-maintainer-mode --disable-dependency-tracking
checking for a BSD-compatible install... /usr/bin/install -c
checking whether build environment is sane... yes
checking for a thread-safe mkdir -p... /usr/bin/mkdir -p
checking for gawk... no
checking for mawk... mawk
checking whether make sets $(MAKE)... yes
checking whether make supports nested variables... yes
checking for gcc... gcc
checking whether the C compiler works... yes
```

```
checking for wchar.h... yes
checking for features.h... yes
checking for wctype.h... yes
checking for getopt.h... (cached) yes
checking for getopt_long_only... yes
checking whether getopt is POSIX compatible... yes
checking for working GNU getopt function... yes
checking for working GNU getopt_long function... yes
checking whether getenv is declared... yes
checking for nl_langinfo and CODESET... yes
checking for symlink... yes
checking for mbsinit... yes
checking for mbrtowc... yes
checking for mbsrtowcs... yes
checking for mprotect... yes
checking for _set_invalid_parameter_handler... no
checking for strndup... yes
checking for iswcntrl... yes
checking for working fcntl.h... yes
checking whether getc_unlocked is declared... yes
checking whether we are using the GNU C Library >= 2.1 or uClibc... yes
checking for a sed that does not truncate output... /usr/bin/sed
checking whether malloc, realloc, calloc are POSIX compliant... yes
checking for mbstate_t... yes
checking for a traditional japanese locale... none
checking for a transitional chinese locale... none
checking for a french Unicode locale... none
checking for a traditional french locale... none
checking for mmap... yes
checking for MAP_ANONYMOUS... yes
checking whether memchr works... yes
checking for stdbool.h that conforms to C99... yes
checking for _Bool... yes
checking for wchar_t... yes
checking whether strerror(0) succeeds... yes
checking for C/C++ restrict keyword... __restrict
checking whether ffsll is declared without a macro... yes
checking whether ffsll is declared without a macro... yes
checking whether memmem is declared without a macro... yes
```

```

Normalized debian/hello/usr/share/locale/ja/LC_MESSAGES/hello.mo
Normalized debian/hello/usr/share/locale/vi/LC_MESSAGES/hello.mo
Normalized debian/hello/usr/share/locale/hr/LC_MESSAGES/hello.mo
Normalized debian/hello/usr/share/locale/ms/LC_MESSAGES/hello.mo
dh_compress
dh_compress: warning: Compatibility levels before 10 are deprecated (level 9 in use)
dh_fixperms
dh_missing
dh_missing: warning: Compatibility levels before 10 are deprecated (level 9 in use)
dh_strip
dh_strip: warning: Compatibility levels before 10 are deprecated (level 9 in use)
877df79ee73ea4e7f8717fa44b1944410a57cb91
dh_makeshlibs
dh_makeshlibs: warning: Compatibility levels before 10 are deprecated (level 9 in use)
dh_shlibdeps
dh_shlibdeps: warning: Compatibility levels before 10 are deprecated (level 9 in use)
dh_installdeb
dh_installdeb: warning: Compatibility levels before 10 are deprecated (level 9 in use)
dh_gencontrol
dh_md5sums
debian/rules override_dh_builddeb
make[1]: Wejście do katalogu '/home/pawel/hello-2.10'
Kosz lddeb -- -Zzstd
dpkg-deb: budowanie pakietu "hello" w "../hello_2.10-2ubuntu4_amd64.deb".
dpkg-deb: budowanie pakietu "hello-dbgsym" w "debian/.debhelper/scratch-space/build-hello/hello-dbgsym_2.10-2ubuntu4_amd64.deb".
Renaming hello-dbgsym_2.10-2ubuntu4_amd64.deb to hello-dbgsym_2.10-2ubuntu4_amd64.ddeb
make[1]: Opuszczenie katalogu '/home/pawel/hello-2.10'
dpkg-genbuildinfo --build=binary -O../hello_2.10-2ubuntu4_amd64.buildinfo
dpkg-genchanges --build=binary -O../hello_2.10-2ubuntu4_amd64.changes
dpkg-genchanges: info: wydanie tylko binarne (bez żadnych źródeł)
dpkg-source --after-build .
dpkg-buildpackage: info: wydanie tylko binarne (bez żadnych źródeł)
Now running lintian hello_2.10-2ubuntu4_amd64.changes ...
running with root privileges is not recommended!
W: hello-dbgsym: elf-error In program headers: Unable to find program interpreter name [usr/lib/debug/.build-id/87/7df79ee73ea4e7f8717fa44b1944410a57cb91.debug]
Finished running lintian.
pawel@myubuntu:~/hello-2.10$

```

Następnie należy przejść do katalogu nadrzędnego i sprawdzić, czy został utworzony pakiet .deb.

```

pawel@myubuntu:~$ ls -al *.deb
-rw-r--r-- 1 root root 53368 maj 31 13:06 hello_2.10-2ubuntu4_amd64.deb

```

Kolejnym krokiem jest instalacja utworzonego pakietu, za pomocą polecenia `sudo dpkg -i hello_*.deb`.

```

pawel@myubuntu:~$ sudo dpkg -i hello_*.deb
Wybieranie wcześniej niewybranego pakietu hello.
(Odczytywanie bazy danych ... 223891 plików i katalogów obecnie zainstalowanych.)
Przygotowywanie do rozpakowania pakietu hello_2.10-2ubuntu4_amd64.deb ...
Rozpakowywanie pakietu hello (2.10-2ubuntu4) ...
Konfigurowanie pakietu hello (2.10-2ubuntu4) ...
Przetwarzanie wyzwalaczy pakietu install-info (6.8-4build1)...
Przetwarzanie wyzwalaczy pakietu man-db (2.10.2-1)...

```

Na końcu można uruchomić program:

```

pawel@myubuntu:~$ hello
Witaj, świecie!

```

Wnioski:

Budowanie pakietu przebiegło bardzo sprawnie, prosto i przyjemnie.